THE PREIMAGE OF A SCHEME THEORETICALLY DENSE OPEN UNDER A FLAT, LOCALLY FINITELY PRESENTED, LOCALLY QUASI-FINITE MORPHISM IS SCHEME THEORETICALLY DENSE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is about morphisms of schemes, and its proof is likewise scheme-theoretic; it is a chapter tool used later for algebraic spaces and is counted as such.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Characterization of scheme theoretic density by functions vanishing on the open. Morphisms, Lemma morphisms-lemma-characterize-scheme-theoretically-dense, used at proof lines 3432-3436. Reduces density to the statement that such a function g must vanish.
- The go-down-with-annihilators lemma for flat locally finitely presented locally quasi-finite morphisms. More on Morphisms, Lemma more-morphisms-lemma-go-down-with-annihilators, used at proof lines 3437-3450. Produces U, the subsheaf F, the integer r and the injection carrying g.
- Scheme theoretic density of W in S. used at proof lines 3452-3454. Forces the sections g_i to vanish.

Source reference. Original lemma 082V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10265–10265. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 01RB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2296–2296. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01RE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2299–2299. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let S be a scheme. Let $W \subset S$ be a scheme theoretically dense open subscheme (Morphisms, Definition 01RB). Let $f : X \to S$ be a morphism of schemes which is flat, locally of finite presentation, and locally quasi-finite. Then $f^{-1}(W)$ is scheme theoretically dense in X.*

Proof. We will use the characterization of Morphisms, Lemma 01RE. Assume $V \subset X$ is an open and $g \in \Gamma(V, \mathcal{O}_V)$ is a function which restricts to zero on $f^{-1}(W) \cap V$. We have to show that $g = 0$. Assume $g \neq 0$ to get a contradiction. By More on Morphisms, Lemma 082V we may shrink V, find an open $U \subset S$ fitting into a commutative diagram

$$\begin{array}{ccc} V & \longrightarrow & X \\ {\scriptstyle \pi}\downarrow & & \downarrow{\scriptstyle f} \\ U & \longrightarrow & S, \end{array}$$

a quasi-coherent subsheaf $\mathcal{F} \subset \mathcal{O}_U$, an integer $r > 0$, and an injective $\mathcal{O}_U$-module map $\mathcal{F}^{\oplus r} \to \pi_*\mathcal{O}_V$ whose image contains $g|_V$. Say $(g_1, \ldots, g_r) \in \Gamma(U, \mathcal{F}^{\oplus r})$ maps to g. Then we see that $g_i|_{W \cap U} = 0$ because $g|_{f^{-1}W \cap V} = 0$. Hence $g_i = 0$ because $\mathcal{F} \subset \mathcal{O}_U$ and W is scheme theoretically dense in S. This implies $g = 0$ which is the desired contradiction. $\square$